Flávio Sousa

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London
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Full-Stack
Algorithms
Blockchain
REPL
Figma
AWS

Toolbox

Clojure(script)
Javascript
Python
C
CUDA/GPU
Node.js
Lua (NGINX)
Tailwind
Postgres
Kafka
Git
Linux

Languages

Spanish (Native)
English (Advanced)

Full-stack developer proficient in Clojure and ClojureScript. Mostly, I am a technologist who enjoys the process of building something with a real-world use case.

I began my career at a fluids' simulation lab, where I published a novel algorithm for forest fire simulation, optimized for GPUs. In 2013 I joined the booming Lisbon startup scene and moved to London in 2015. I have hacked code for client demos and been a developer advocate for early stage startups, and also enjoyed working on well financed greenfield re-architectures. I believe people solve problems not tools, so I always look for the most effective solution. I tend to prioritize languages over frameworks, and I can write infrastructure as code when necessary.

I set up Codecadre LTD for my contracting work in 2019. Besides software contracting, I'm available to work with NGOs and offer free mentoring for people from a low-income background.

Education

Msc. in Mechanical Engineering at University of Lisbon Sep 2008 - Jun 2011
Course representative 2009/2010. Worker student. Grade: A.

Bsc. in Mechanical Engineering at University of Lisbon Sep 2004 - Sep 2008
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Experience

Independent freelancer
Remote

Between contracts I made several contributions under CodeCadre LTD.

- Maintaining a free dataset and web UI of driving school pass rates.
- Acting as a tech advisor for multiple early-stage businesses.
- Avid learner - acquired multiple Coursera certificates on various topics.
- Mentoring individuals from low-income backgrounds.
- Speaking at tech conferences.

CLOJURE AND CLOJURESCRIPT - BABASHKA - TAILWIND - CRYPTOGRAPHY

Clojure contractor at Awaze Aug 2022 - Dec 2022
Manchester

As a Clojure contractor for Awaze, I was placed in a team that was implementing a unified search API across multiple holiday rental businesses. All code was produced alongside the team while "mobbing".

- Spoke about the workflow methodology at a conference
- Made several critical contributions that helped speed up delivery
- Mentored new members of the team and conducted regular Friday afternoon Clojure practice sessions
- Successfully Identified bottlenecks with a custom lightweight tool
- Taught Clojure best practices to the rest of the team

CLOJURE - AWS - POSTGRES - KOTLIN

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Languages

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Experience

Full-Stack developer at Momentum AI

Aug 2019 - Aug 2021

Colorado (Remote)

- Made a significant contribution in helping the initial consulting business transition into a successful remote sensing company.
- Frontend improvements - Drastically reduced the frontend build time by migrating to shadow-cljs and moving to a component-based architecture with devcards.
- Refactored the existing data model to enhance query efficiency of the timeseries sensor data.
- Fully owned a critical Elastic Search instance that monitored and alerted on the remote sensors in production.

CLOJURE AND CLOJURESCRIPT - REAGENT - RE-FRAME - POSTGRES - TIMESCALE - ELASTICSEARCH

Cloud Services Engineer at Youview

Oct 2018 - Aug 2019

London

- As a valuable member of the Youview team, contributed to the set-top box provider for BT and TalkTalk, writing software in-house and serving requests for 2.6 million clients.
- Successfully reduced the AWS bill by responsibly consolidating a set of microservices, thereby optimizing the cloud infrastructure and reducing overall costs.
- Proactively identified and advocated for the refactoring of an overly fragmented set of microservices to improve efficiency,streamlining processes and enhancing overall service quality.

NODE.JS - CLOJURE - AWS - MICROSERVICES

Clojure Engineer at Funding Circle

May 2017 - Sep 2018

London

- Played a key role as part of the in-house team that migrated a legacy Ruby platform to a modern Clojure and Kafka stack at Funding Circle, enabling more efficient business loans for private and institutional investors.
- Volunteered as part of a crisis team for production issues on multiple occasions, demonstrating excellent problem-solving skills under pressure.
- Developed effective in-house tools to manage Kafka connections in development environments, contributing to the overall efficiency of the work processes.
- Collaborated remotely with teams located in Berlin and San Francisco, demonstrating effective communication and teamwork skills.

CLOJURE - KAFKA - DOCKER - RUBY - MICROSERVICES

Software Engineer at Style.com

Apr 2015 - May 2017

London

- Worked collaboratively with a team of 30+ engineers and various contract partners to develop a green field e-commerce marketplace for Condé Naste. The marketplace aggregated data feeds from several brands with AI curation.
- Proactively took the initiative to learn new languages, including Clojure and Lua
- Released several stages of the online store, delivering high-value solutions to our clients.
- Contributed significantly by designing and implementing an Nginx/Lua solution to ease the pre-launch process and improve infrastructure.
- Worked with different teams including the API team, marketing team,and backend services team, specifically with ingesting product feeds.

CLOJURE/CLOJURESCRIPT - NODE.JS - DOCKER - NGINX/LUA/OPENRESTY

Independent Full-Stack freelancer

Mar 2014 - Apr 2015

Lisbon

- Was responsible for finding my own clients and closing deals.

- Oversaw all stages of the projects, including gathering requirements, presenting quotes, development, deployment, and infrastructure management.
- Contributed to the early stage launch of products, which are still live as of 2023.
- Worked with a diverse range of cutting-edge technologies.

RUBY ON RAILS - NODE.JS - SHOPIFY

Application Developer at Crowdprocess

Mar 2013 - Mar 2014

Lisbon

- As the third employee of an early-stage startup, played a pivotal role in conducting research and development of applications for a distributed computing platform.
- Provided exceptional support and documentation to clients, ensuring that all needs were met and exceeded expectations.
- Worked collaboratively with experts to implement cutting-edge algorithms across multiple scientific fields such as computer vision, neuroscience, bioinformatics, genetic algorithms, and forest fire prediction.
- Prominent speaker at conferences and tech meetups, sharing my insights on the latest industry trends and advancements.
- Developed an innovative forest-fire prediction software, based on original research, which has been highly acclaimed by experts in the field.

JAVASCRIPT - NODE.JS - POSTGRES - C/C++ - PYTHON - AWS

Research Assistant at University of Lisbon

Jan 2010 - Nov 2012

Lisbon

- During my Master's coursework, I conducted basic research in a fluid simulation lab and earned a research grant for my dissertation.
- Achieved speed-ups of 200 times by porting an open-source forest fire library written in C to GPUs.
- Co-authored two papers with 28 citations as of 2023.
- In September 2010, I secured an FCT research grant to conduct CFD analysis of high-speed trains and renewable energy cogeneration devices.

C - CUDA - JAVA - STARCCM+ - PARAVIEW

Research Papers

- Faster Than Real Time Stochastic Fire Spread Simulations, 2013. doi
- Simulation of surface fire fronts using firelib and GPUs, 2012. doi
- Numerical Investigations on the Inter-Car-Gap Flow for a Generic High Speed Train, 2011. doi

Awards

- Recognized as 2nd Place winner in the MSc. Category of the 2011 Fraunhofer Challenge, awarded for a practical research project.
- As a 2011 ISCTE-IUL MIT Startup Pitch Competition semi-finalist, contributed by performing a fluid flow simulation for a high altitude wind turbine project.